

← Investment Theory

REVIEW

Summary Review Notes

Condensed review of all chapters -- key concepts, equations, and takeaways for exam preparation.

Chapter I: Capital Markets and Investment Performance

Core Topics

Geometric vs. arithmetic returns -- Geometric returns reflect actual compound growth; arithmetic returns are useful for statistical modeling. Historical capital market data (Ibbotson Associates, 1926-1995). The **equity premium** averaged approximately 8.5% (arithmetic). Standard deviation as the primary measure of volatility.

BASIC RETURN FORMULA

$$R_t = \frac{P_{t+1} - P_t + D_t}{P_t}$$

Chapter II: The Efficient Frontier

Core Topics

Risk and return -- the fundamental trade-off. Correlation between assets determines diversification benefits. The **efficient frontier** is the set of portfolios offering maximum

return for each level of risk. Adding a riskless asset creates the Capital Allocation Line.

← Investment Theory

PORTFOLIO VARIANCE (TWO ASSETS)

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Chapter III: Investor Preferences and the Tangency Portfolio

Core Topics

Iso-utility curves represent investor preferences over risk and return. The **shortfall approach** and **Sharpe ratio** provide practical criteria for portfolio selection. The **tangency portfolio** is the optimal risky portfolio -- the point where the Capital Allocation Line is tangent to the efficient frontier.

SHARPE RATIO

$$S = \frac{E[R_p] - R_f}{\sigma_p}$$

Chapter IV: The Capital Asset Pricing Model

Core Topics

The **tangency portfolio equals the market portfolio** in equilibrium (all investors hold the same risky portfolio). The **CAPM equilibrium**: expected return is a linear function of beta. The **"Birthday Cake" proof** demonstrates that the market portfolio must be the tangency portfolio when all investors hold mean-variance optimal portfolios.

← Investment Theory

$$E[R_i] = R_f + \beta_i (E[R_m] - R_f)$$

Chapter V: Further Explorations of the CAPM

Core Topics

Correlations drive attractiveness of a stock -- low correlation with the market portfolio means the asset shifts the frontier favorably. The **Security Market Line** plots expected return against beta. Beta is the sole risk measure: $\beta = \text{cov}(R_i, R_m) / \text{var}(R_m)$. Only systematic risk is priced.

BETA

$$\beta_i = \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)} = \frac{\rho_{i,m} \sigma_i}{\sigma_m}$$

Chapter VI: The Arbitrage Pricing Theory

Core Topics

The **APT** allows multiple sources of systematic risk. **Arbitrage in expectations**: if a security deviates from the SML, investors exploit the mispricing by going long the underpriced asset and shorting a same-beta portfolio. Multi-factor model: $E[R_i] = R_f + \sum_k \beta_{i,k} \lambda_k$. Chen, Roll, and Ross identified macro factors: inflation, GNP, investor confidence, yield curve shifts.

Chapter VII: Where Do Betas Come From?

← Investment Theory

Beta estimation via regression of security returns on market returns (typically 60 months). **Leverage increases equity beta:** $\beta_{\text{equity}} = \beta_{\text{asset}} \times (D + E)/E$. **Unlevering** to find asset beta from comparable firms, then **relevering** to your own capital structure. Application to **project valuation, M&A, and capital budgeting** -- each project should use its own beta, not the company-wide cost of capital.

UNLEVERING / RELEVERING

$$\beta_{\text{asset}} = \frac{E}{D + E} \beta_{\text{equity}} \quad \Longleftrightarrow \quad \beta_{\text{equity}} = \beta_{\text{asset}} \times \frac{D + E}{E}$$

Chapter VIII: Information and Market Efficiency

Core Topics

Market efficiency: how rapidly information embeds in prices. Three forms: **Weak** (past prices), **Semi-Strong** (all public info), **Strong** (all info including private). Evidence: event studies show rapid adjustment; professional managers show no consistent outperformance; daily momentum exists but is not profitably exploitable. **Random walk** approximation holds for liquid markets. Efficient markets benefit investors through price stability, reduced search costs, and liquidity.

Chapter IX: Long-Term Corporate Financing

Core Topics

Two primary approaches: **Selling ownership** (stocks, warrants) and **Borrowing** (bonds, preferred shares). The **life cycle of stock**: design, IPO, exchange trading. Bond features: coupon, maturity, covenants, callability, sinking funds. **Preferred shares:**

← Investment Theory

Key Equations Reference

CONCEPT	EQUATION
Return	$R_t = (P_{t+1} - P_t + D_t)/P_t$
Portfolio variance	$\sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_{ij}$
Sharpe ratio	$S = (E[R_p] - R_f)/\sigma_p$
CAPM	$E[R_i] = R_f + \beta_i(E[R_m] - R_f)$
Beta	$\beta_i = \text{cov}(R_i, R_m)/\text{var}(R_m)$
APT (K factors)	$E[R_i] = R_f + \sum_k \beta_{i,k} \lambda_k$
Unlever beta	$\beta_{\text{asset}} = [E/(D + E)] \beta_{\text{equity}}$
Growing perpetuity	$P = \text{CF}/(R - g)$
Discount rate from P/E	$R = 1/(P/E) + g$

Acknowledgement: The interactive tools referenced in this textbook are from the [Rice Business Investments Compendium](#), created and maintained by Kerry Back and Kevin Crotty at the Jones Graduate School of Business, Rice University. We gratefully acknowledge their contribution to investment education.

← Investment Theory

← Chapter IX: Long-Term Corporate Financing

Chapter I: Capital Markets →

© William N. Goetzmann, Yale School of Management